Kedros Analytics

MARKET BASKET ANALYSIS

Know What Sells Together

Your sales data already shows which products belong together.
Turn it into cross-sells, bundles and smarter stock decisions.

Company Page
Blog

Matt Protopapas, PhD

THE OPPORTUNITY

Every checkout leaves evidence. Most SMEs never look at it

Your point-of-sale or e-commerce system already records which products are bought together. It just sits there, unused.

GUT FEEL

“Chips and beer go together, let’s put them next to each other.”

Often right, but it only covers the few pairs one person happens to think of. The non-obvious combinations stay invisible.

MARKET BASKET ANALYSIS

Every combination, every time.

A few hundred products and a few thousand sales is too much to spot by eye, yet trivial for a computer to mine completely.

WHAT IT IS

From “we think these sell well together” to “we know, and by how much”

Market basket analysis asks which items get bought together, and how strongly. It is the sales-side companion to inventory optimization.

1

Your sales data

Export 3–6 months of transactions from your POS or e-commerce platform. Most systems can produce a CSV.

→

2

Analyze every combination

The Apriori algorithm scans every possible product combination across all baskets.

→

3

Ranked, proven rules

Each rule comes with support, confidence and lift, so you know which ones are worth acting on.

Demo store: 3,000 baskets · 30 products · 71 association rules found automatically

Demo figures from a synthetic six-month grocery dataset, for illustration only.

WHAT YOU GAIN

Six ways it pays off for a small business

1 Increase sales through cross-selling

“Frequently bought together” prompts based on evidence, not guesswork.

2 Improve product placement

Complementary items placed together, in-store or on a product page, capture impulse purchases at no marketing cost.

3 Sharpen promotions and bundling

Bundling items already bought together protects margin better than discounting each one.

4 Feed back into inventory

A top seller’s companion product has predictable demand too, which links directly to reorder-point thinking.

5 Personalize marketing for free

Recommend the item a customer is statistically likely to want next in email or SMS.

6 Inform assortment decisions

Customers buy pasta but you stock no matching sauce? That is a visible, quantified gap.

Fastest wins for most SMEs: cross-selling and bundling (highlighted). Both use data you already have and cost nothing to test.

THE THREE NUMBERS

Support, confidence and lift keep this honest

Worked example from the demo store: the rule “Butter + Jam → Bread”.

SUPPORT

2.4%

How common is the combination across all baskets?

Roughly 1 in 40 baskets. Low support means a real but niche pattern.

CONFIDENCE

100%

Of baskets with the first item(s), how many also hold the second?

Every butter-and-jam basket also had bread: a reliable prediction.

LIFT

10.9×

Is the link real, or just because the item is popular anyway?

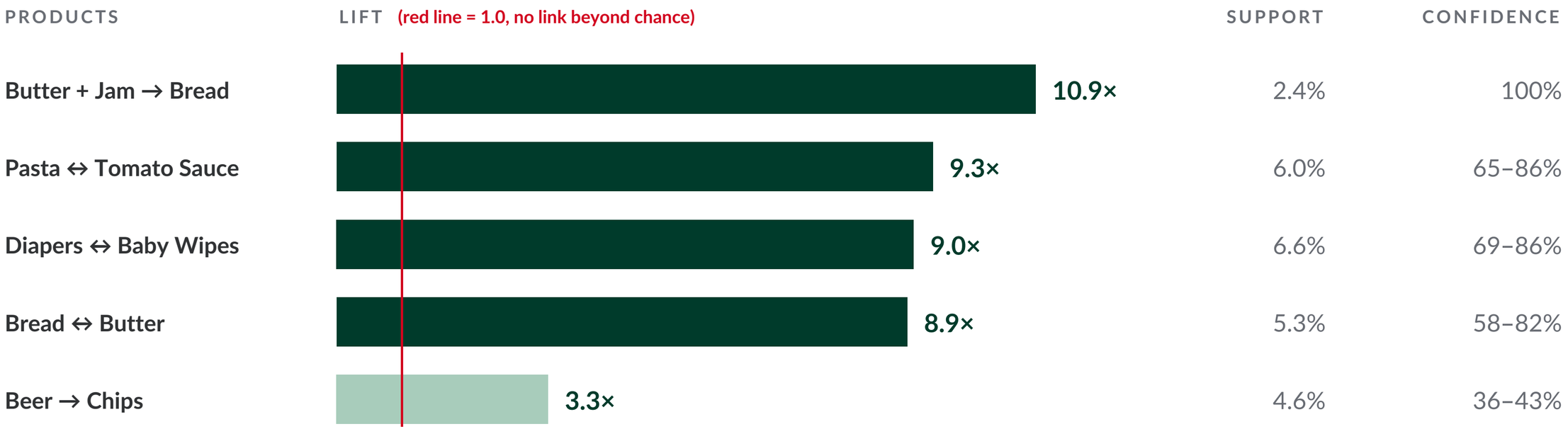
Bread is almost 11 times more likely in these baskets than in an average one.

Rule of thumb: lift above 1 is a real association, and higher is stronger. Always sanity-check it against support.

EXAMPLE RESULTS

Strong patterns, found automatically

Top five of 71 rules, ranked by lift. A lift of 1 would mean no link beyond chance.



Bread and butter are intuitive. Diapers and wipes also pull beer along more than average, a pattern nobody would think to check.

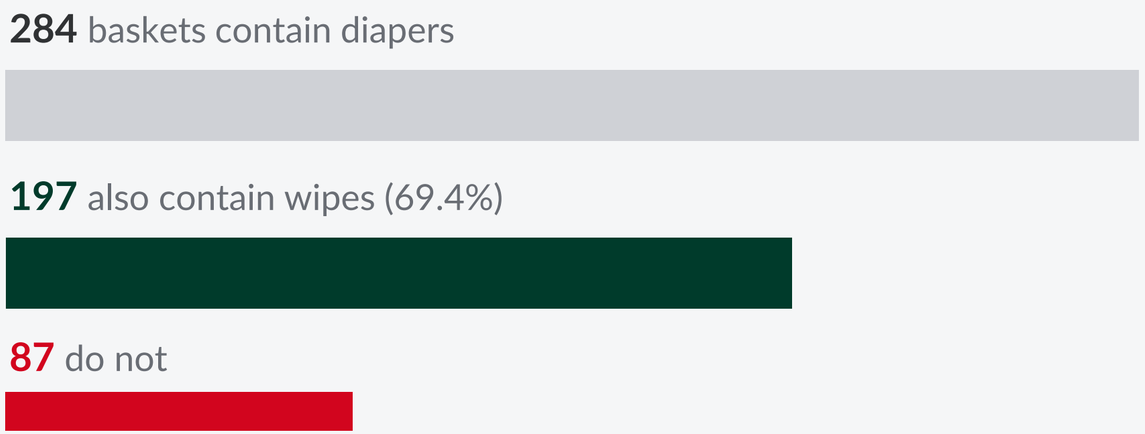
Synthetic demo data (3,000 baskets, 30 products). Confidence ranges reflect the two directions of each rule.

BUSINESS IMPACT

Even the strongest patterns leave money on the table

Diapers and baby wipes have a lift of 9.0, yet nearly a third of diaper buyers still skip the wipes.

SIX MONTHS OF DEMO-STORE BASKETS



THE CEILING IS ACHIEVABLE

85.7%

of wipe buyers already pick up diapers.

Closing half the gap (69% → ~78%) means about 23 extra wipe sales, roughly $115 in six months.

Synthetic demo data; wipes at $4.99 a pack. Small on its own, but see the next slide.

WHY IT COMPOUNDS

Free to act on, repeated across the whole catalog

Modest in the demo, bigger in practice, for three reasons.

01

Real SMEs have more volume

Scale the demo's transactions up 5–10× (still a local shop, not a chain) and the same percentage-point gain scales with it.

02

This is one rule out of 71

Bread/butter/jam, pasta/sauce, cereal/milk and beer/chips/salsa are separate, simultaneous opportunities. The value is dozens of small nudges.

03

The cost of acting is near zero

Moving a shelf, adding a "goes well with" line online or printing a receipt suggestion costs far less than a discount or an ad campaign.

GET STARTED

What to do this week

Five steps from raw sales data to a measurable test.

1

Export

Last 3–6 months of transactions from your POS or e-commerce platform.

2

Reshape

One row per product per receipt, then run it through the analysis pipeline.

3

Rank

Sort by lift and ignore anything with support below roughly 1–2%.

4

Act

Pick the top 3–5 rules with a good-margin item being pulled along. Make the pairing visible for 30 days.

5

Repeat

Re-run next quarter. Buying patterns drift with seasons and promotions.

Step 3 matters: a lift of 20 on a combination in only 3 of 3,000 baskets is real but too rare to build a promotion around.

KEEP IT HONEST

The limits worth keeping in mind

The output is an input to a merchandising or marketing decision, not the decision itself.

1

Association, not causation

It shows what appears together, not why.

2

Needs clean data

Messy exports are common. A data-quality gate (20 checks in the demo) catches problems before they skew results.

3

Needs real volume

A handful of sales won’t produce reliable patterns.

4

Reflects the past

Last year’s holiday rule may not hold today. Re-run periodically.

HOW KEDROS ANALYTICS CAN HELP

Try it on your own data, with a free proof of concept

Business questions first, AI as a tool and not a hype machine, and results you can act on straight away.

1

Validate

Check your sales export against data-quality rules so the analysis starts from data you can trust.

2

Mine and rank

Find the association rules in your own baskets and rank them by lift, support and confidence.

3

Act and track

Choose the 3–5 rules worth testing, make them visible, and monitor the gains over time.

Free PoC: tailored to your requirements and your company’s analytics maturity.

Kedros Analytics

Ready to see what your sales reveal?

Let's spend thirty minutes on your sales data. No commitment, no jargon, and we might even plan a free PoC.

Book a free call

Matt@KedrosAnalytics.com



Company Page
Blog

Matt Protopapas, PhD · Founder